

# Experiment #1 – Clock and Periodic Signal Generation

Students name: Amirali Hajizadeh Moghaddam – Bahareh Bakhtiari  
810102430 - 810102406

**Abstract**— In this experiment, different methods of clock and periodic signal generation were studied and implemented, including a ring oscillator, an LM555 timer in astable mode, and frequency division using synchronous counters. The oscillation frequencies were measured and compared with theoretical calculations. The experimental results showed close agreement with expected values, demonstrating the relationship between resistance, capacitance, and output frequency in oscillatory circuits.

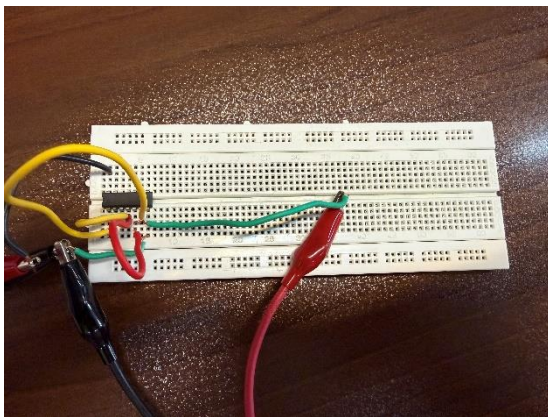
**Keywords**— Clock generation, Ring oscillator, LM555 timer, Frequency divider, T flip-flop

## I. INTRODUCTION

THE PURPOSE OF THIS EXPERIMENT IS TO UNDERSTAND THE GENERATION OF PERIODIC CLOCK SIGNALS USING ANALOG AND DIGITAL COMPONENTS. THE CIRCUITS INCLUDE RING OSCILLATORS IMPLEMENTED BY 74HCT04 INVERTERS, THE LM555 TIMER IN ASTABLE MODE, AND SYNCHRONOUS COUNTERS FOR FREQUENCY DIVISION. THESE CIRCUITS ARE ESSENTIAL BUILDING BLOCKS FOR DIGITAL SYSTEM TIMING AND CONTROL.

## II. EXPERIMENTAL SETUP AND PROCEDURE

### A. RING OSCILLATOR



THE CIRCUIT WAS IMPLEMENTED USING FIVE INVERTERS OF THE 74HCT04 IC CONNECTED IN SERIES. THE OUTPUT OF THE LAST INVERTER WAS FED BACK TO THE INPUT OF THE FIRST ONE. THE MEASURED OSCILLATION FREQUENCIES RANGED BETWEEN 1.35 MHz AND 8.98 MHz AS SHOWN IN FIGURES 1–5.



FIGURE 1 – WITHOUT RESISTANCE (FREQ = 8.98 MHz)

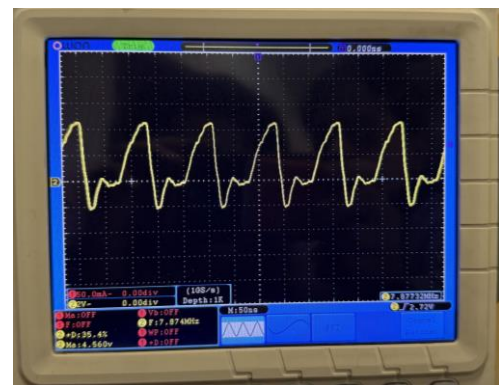


FIGURE 2 - RESISTANCE = 1KOHM (FREQ = 7.87 MHz)

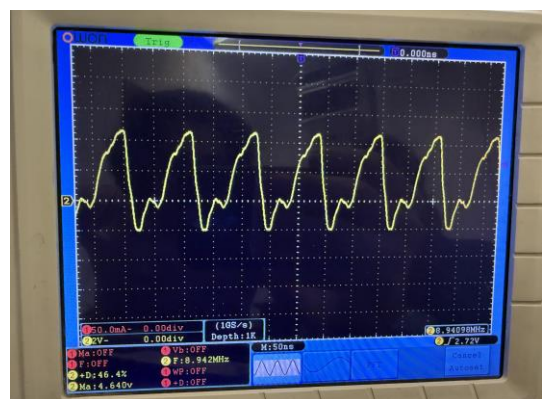
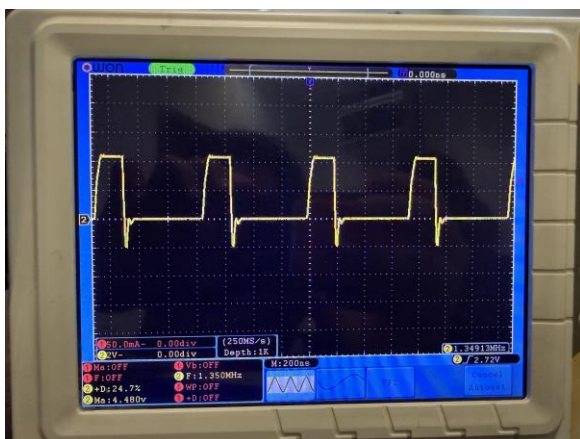
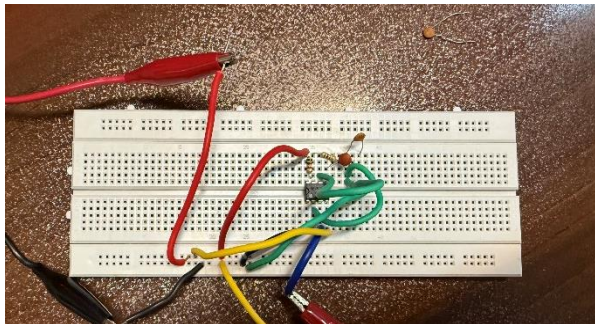


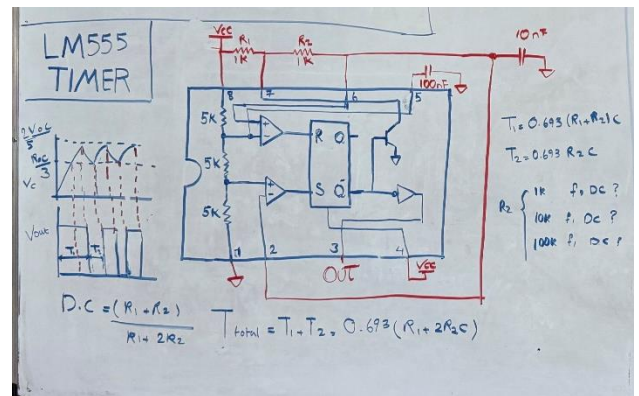
FIGURE 3 - RESISTANCE = 100 OHM (FREQ = 8.94 MHz)



## B. LM555



THE LM555 TIMER WAS CONNECTED AS SHOWN IN FIGURE 7 (SCHEMATIC ON THE WHITEBOARD). USING  $R1 = 1\text{ k}\Omega$ ,  $R2 = \{1\text{ k}\Omega, 10\text{ k}\Omega, 100\text{ k}\Omega\}$ , AND  $C = 10\text{ nF}$ , THE OSCILLATION FREQUENCIES WERE MEASURED AND COMPARED WITH THEORETICAL CALCULATIONS. FIGURE 6–10.





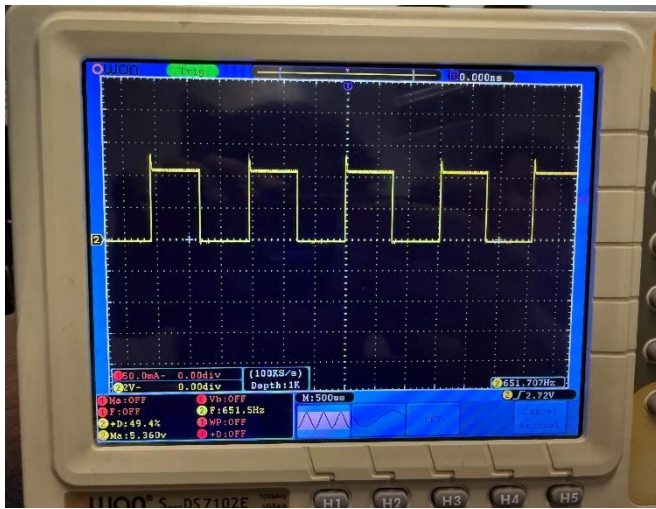


FIGURE 10 - RESITANSE = 100 KOHM - DUTYCYCLE = 49.7%

### C. SYNCHRONOUS COUNTER AS A FREQUENCY DIVIDER

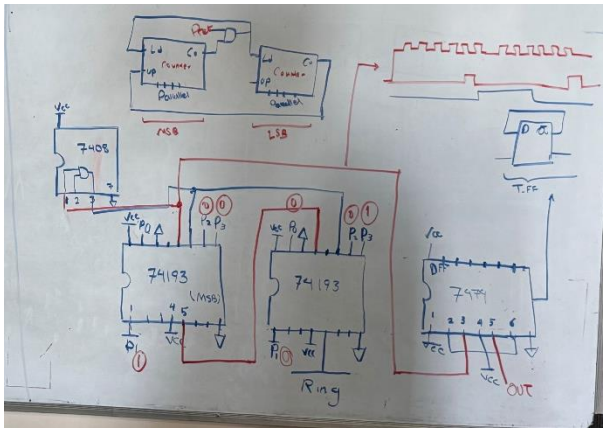


FIGURE 11 - FREQUENCY DIVIDER STRUCTURE

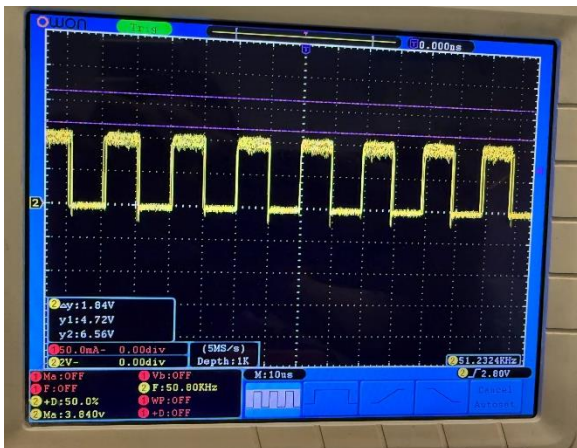


FIGURE 12 - FREQUENCY = 50.8 KHZ

## III. RESULTS AND DISCUSSION

### PART A)

- i. The oscillation frequency of the ring oscillator was measured to be 8.98 MHz. The corresponding time periods are:

$$T = 1 \div f \text{ then } T = 111.35 \text{ ns.}$$

Propagation Delay is the time between 50% point of input to the 50% point of output. So the propagation delay is 55.67 ns.

- ii. Delay of a Single Inverter: Since the period of oscillation in a ring oscillator is related to the number of inverters and the delay of each inverter then the delay of an inverter is:

$$T_{inv} = T \div N \text{ then } T_{inv} = 55.67 \div 3 = 18.55 \text{ ns}$$

- iii. By adding a resistor and capacitor to the feedback path the oscillation frequency decreased significantly. This demonstrates that the **RC network increases the total delay** in the feedback loop, resulting in a lower oscillation frequency.

### PART B)

$$T_1 = 0.693 * (R_1 + R_2) * C$$

$$T_2 = 0.693 * R_2 * C$$

$$T = T_1 + T_2 = 0.693 * (R_1 + 2R_2) * C$$

CALCULATION ABOUT FIGURE 8 :

$$T_1 = 0.693(1 + 1) \times 10^3 \times 10^{-8} = 13.86 \mu\text{s}$$

$$T_2 = 0.693 \times 1 \times 10^3 \times 10^{-8} = 6.93 \mu\text{s}$$

$$T = T_1 + T_2 = 13.86 + 6.93 = 20.79 \mu\text{s}$$

$$\text{DutyCylce} = \frac{T_1}{T} = \frac{13.86}{20.79} = 0.66 = 66\%$$

$$\text{Practical Value of DutyCycle} = 65\%$$

So it is correct

CALCULATION ABOUT FIGURE 9 :

$$T_1 = 0.693(1 + 10) \times 10^3 \times 10^{-8} = 76.23 \mu\text{s}$$

$$T_2 = 0.693 \times 10 \times 10^3 \times 10^{-8} = 69.3 \mu\text{s}$$

$$T = T_1 + T_2 = 76.23 + 69.3 = 145.53 \mu\text{s}$$

$$\text{DutyCylce} = \frac{T_1}{T} = \frac{76.23}{145.53} = 0.5238 = 52.38\%$$

$$\text{Practical Value of DutyCycle} = 51.6\%$$

So it is correct

CALCULATION ABOUT FIGURE 10 :

$$T_1 = 0.693(1 + 100) \times 10^3 \times 10^{-8} = 699.93 \mu\text{s}$$

$$T_2 = 0.693 \times 100 \times 10^3 \times 10^{-8} = 693 \mu\text{s}$$

$$T = T_1 + T_2 = 699.93 + 693 = 1392.93 \mu\text{s}$$

$$\text{DutyCylce} = \frac{T_1}{T} = \frac{699.93}{1392.93} = 0.5025 = 50.25\%$$

$$\text{Practical Value of DutyCycle} = 49.7\%$$

So it is correct

## PART C)

The clock signal of Ring oscillator was connected as input of LSB

The present value = 56

To represent this number we loaded counters parallel input with 56 (binary 00111000)

The T flip flop correctly produced the output with 50 % DutyCycle.

According to figure 12 we can see the frequency is 50.8 KHz .

And also we calculate the theorical value:

$$\frac{8.98}{256 - 56} = 0.0449 \text{ MHz} = 44.9 \text{ KHz}$$

### Explanation of Part C)

It is often difficult to generate very low-frequency signals directly because even a small timing error (about 1–2%) causes a large deviation at low frequencies. To overcome this, synchronous counters such as the 74HC193 can be used to divide a higher-frequency clock down to a lower one.

The 74HC193 is a 4-bit up/down counter with parallel load inputs (P3–P0) that can preset an initial value. For an 8-bit counter made of two cascaded 74HC193s, the initial value is calculated by:

$$\text{Initial Value} = \text{Maximum Value} - \text{Modulus}$$

For a modulus of 200, we load 56 (0x38) as the starting value.

The carry-out of the LSB counter is connected to the clock input of the MSB counter, and the clock input of the LSB comes from the ring oscillator built in Part A. A 7408 AND gate is used to combine the preset signal and the carry-out of the MSB to generate a parallel load pulse whenever one counting cycle is completed.

As a result, the input frequency from the ring oscillator (about 9 MHz) is divided by 200, producing an output signal of approximately 50 kHz, as shown in the oscilloscope results.

### C. T FLIP-FLOP

To obtain a square wave with a 50% duty cycle, a T flip-flop is used after the counter stage.

A T flip-flop toggles its output on each clock pulse, effectively dividing the input frequency by two.

In this experiment, the 74HC74 D-type flip-flop is configured as a T flip-flop by connecting its Q' output to the D input (that is,  $D = \neg Q$ ).

The clock input of this flip-flop is connected to the carry-out signal of the MSB counter from the previous stage.

As a result, the output frequency becomes half of the counter's carry frequency, and the output waveform achieves a nearly perfect 50% duty cycle.

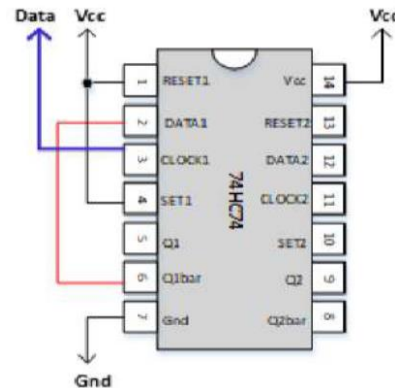


Figure 13 - pin layout of 74HC74

## VI . CONCLUSIONS

The general ideas discussed in this experiment were:

- A. Power supply, Function Generator, and Oscilloscope
- B. 74 Series Basic Logic Gates
- C. Different oscillator circuits (a LM555 timer IC, Schmitt trigger Oscillator)
- D. Sequential circuits using clocks and counters

## ACKNOWLEDGMENT

This report was prepared by **Bahareh Bakhtiari** and **AmirAli Hajizadeh Moghaddam** for 1,2 sessions of DLD Laboratory at ECE department in 1404-1405 fall semester.

## REFERENCES

- [1] Dr. Navabi's slides for DLD course.
- [2] Digital Logic Laboratory Manual, "Clock and Periodic Signal Generation," University of Tehran, School of Electrical and Computer Engineering, Fall 1405.
- [3] Texas Instruments, "LM555 Timer," Datasheet, Rev. SLOS118P, 2021.
- [4] Nexperia, "74HC04 Hex Inverter," Product Datasheet, 74HC\_HCT04, 2022.
- [5] Nexperia, "74HC74 Dual D-type Flip-Flop with Set and Reset," Product Datasheet, 74HC\_HCT74, 2022.
- [6] Texas Instruments, "SN74HC193 4-Bit Synchronous Up/Down Counter," Datasheet, Rev. SCLS085H, 2020.

